Create resource groups for VMs and datastores

SnapCenter Plug-in for VMware vSphere

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Table of Contents

Create resource groups for VMs and datastores 1

Create resource groups for VMs and datastores

A resource group is the container for one or more VMs or datastores that you want to protect.

Resource groups can contain any combination of VMs and datastores. Beginning with SnapCenter Plug-in for VMware vSphere 4.4, a resource group can also contain one FlexGroup volume or one or more FlexVol volumes, but not both.

> For spanning VMs, using FlexGroup and FlexVol together is not supported.

SnapCenter Plug-in for VMware vSphere 4.5 and later supports datastores on large LUN sizes up to 128 TB on ASA aggregates. For large LUNs, SnapCenter uses only thick provisioned LUNs to avoid latency.

About this task

You can add or remove VMs and datastores from a resource group at any time.

- Backing up a single resource

  To back up a single resource (for example, a single VM), you must create a resource group that contains that single resource.

- Backing up multiple resources

  To back up multiple resources, you must create a resource group that contains multiple resources.

- Resource groups that contain FlexGroup volumes in MetroCluster environments

  If you are running on ONTAP 9.8 or ONTAP 9.9, then after a switchover or switchback, you must restart the SnapCenter VMware plug-in service and resynchronize the SnapMirror relationships before you back up resource groups in MetroCluster environments.

  In ONTAP 9.8, the backups hang after the switchback. This issue is fixed in ONTAP 9.9.

- Optimizing Snapshot copies

  To optimize Snapshot copies, you should group the VMs and datastores that are associated with the same volume into one resource group.

- Backup policies

  Although it is possible to create a resource group without a backup policy, you can only perform scheduled data protection operations when at least one policy is attached to the resource group. You can use an existing policy, or you can create a new policy while creating a resource group.

  > A resource group can contain VMs, and SAN and NAS datastores; it cannot contain VSAN or vVOL datastores.

  > Do not add VMs that are in an inaccessible state. Although you can create a resource group that contains inaccessible VMs, backups for that resource group will fail.

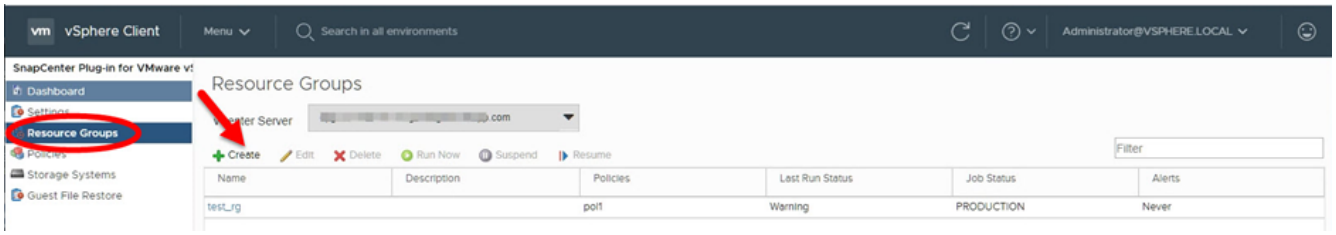
- Compatibility checks

SnapCenter performs compatibility checks when you create a resource group.

Manage compatibility check failures

Steps

1. In the left Navigator pane of the VMware vSphere web client, click **Resource Groups**, then click ➕ **Create** to start the wizard.



This is the easiest way to create a resource group. However, you can also create a resource group with one resource by performing one of the following:

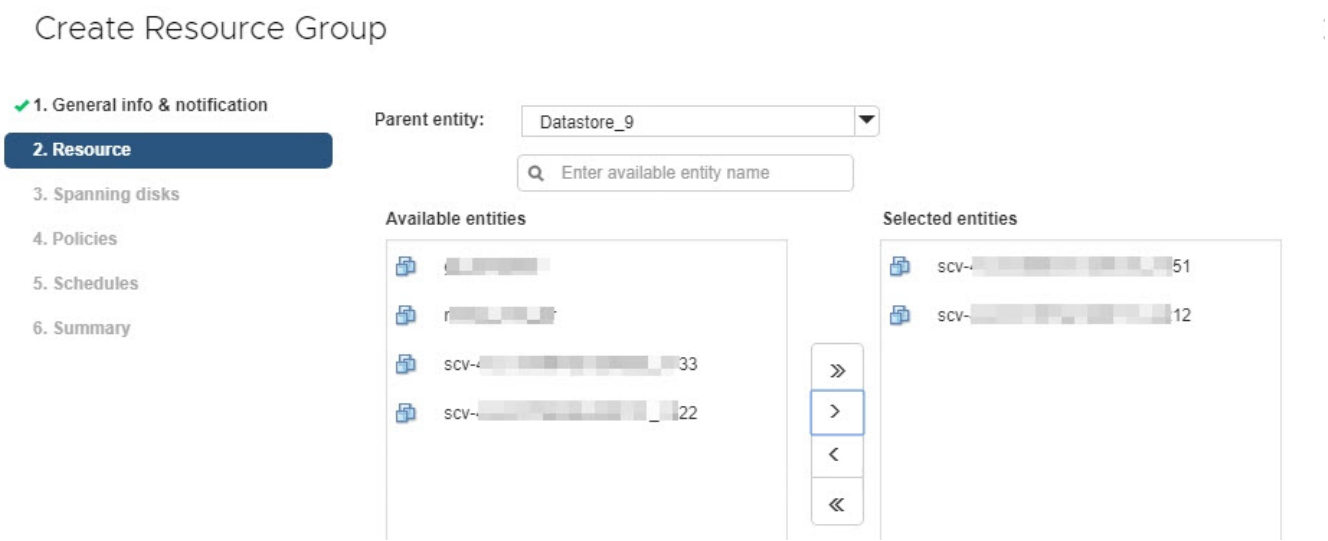
- To create a resource group for one VM, click **Menu** > **Hosts and Clusters**, then right-click a VM, then select NetApp SnapCenter, and then click ➕ **Create Resource Group**.
- To create a resource group for one datastore, click **Menu** > **Hosts and Clusters**, then right-click a datastore, then select **NetApp SnapCenter**, and then click ➕ **Create Resource Group**.

2. On the **General Info & Notification** page in the wizard, do the following:

For this field…	Do this…
vCenter Server	Select a vCenter server.
Name	Enter a name for the resource group. Do not use the following special characters in VM, datastore, policy, backup, or resource group names: % & * $ # @ ! \ / : * ? " < > - [vertical bar] ; ' , . An underscore character (_) is allowed. VM or datastore names with special characters are truncated, which makes it difficult to search for a specific backup. In linked mode, each vCenter has a separate SnapCenter VMware plug-in repository. Therefore, you can use duplicate names across vCenters.
Description	Enter a description of the resource group.
Notification	Select when you want to receive notifications about operations on this resource group: Error or warnings: Send notification for errors and warnings only Errors: Send notification for errors only Always: Send notification for all message types Never: Do not send notification
Email send from	Enter the email address you want the notification sent from.

For this field…	Do this…
Email send to	Enter the email address of the person you want to receive the notification. For multiple recipients, use a comma to separate the email addresses.
Email subject	Enter the subject you want for the notification emails.
Latest Snapshot name	Beginning with SnapCenter Plug-in for VMware vSphere 4.4, if you want the suffix "_recent" added to the latest Snapshot copy, then check this box. The "_recent" suffix replaces the date and timestamp. **NOTE:** A `-recent` backup is created for each policy that is attached to a resource group. Therefore, a resource group with multiple policies will have multiple `-recent` backups.
Custom Snapshot format	If you want to use a custom format for the Snapshot copy names, then check this box and enter the name format. * By default, this feature is disabled. * The default Snapshot copy names use the format `<ResourceGroup>_<Date-TimeStamp>` However, you can specify a custom format using the variables $ResourceGroup, $Policy, $HostName, $ScheduleType, and $CustomText. Use the drop-down list in the custom name field to select which variables you want to use and the order in which they are used. If you select $CustomText, the name format is `<CustomName>_<Date-TimeStamp>`. Enter the custom text in the additional box that is provided. NOTE: If you also select the "_recent" suffix, you must make sure that the custom Snapshot names will be unique in the datastore, therefore, you should add the $ResourceGroup and $Policy variables to the name. * Special characters For special characters in names, follow the same guidelines given for the Name field.

3. On the **Resources** page, in the Available entities list, select the resources you want in the resource group, then click **>** to move your selections to the Selected entities list.



When you click **Next**, the system first checks that SnapCenter manages and is compatible with the storage on which the selected VMs or datastores are located.

If the message `Selected virtual machine is not SnapCenter compatible or Selected datastore is not SnapCenter compatible` is displayed, then a selected VM or datastore is not compatible with SnapCenter. See Manage compatibility check failures for more information.

4. On the **Spanning disks** page, select an option for VMs with multiple VMDKs across multiple datastores:

   Always exclude all spanning datastores [This is the default for datastores.]

   Always include all spanning datastores [This is the default for VMs.]

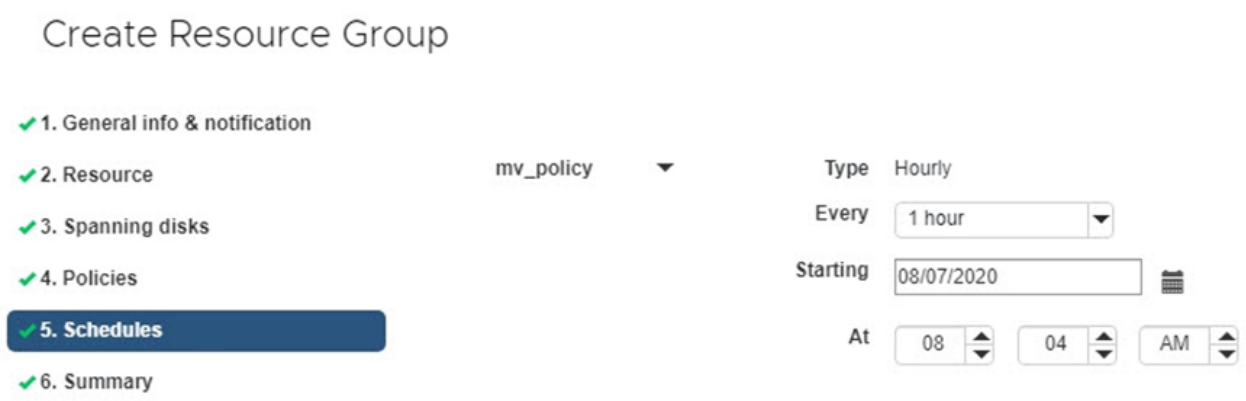
   Manually select the spanning datastores to be included

5. On the **Policies** page, select or create one or more backup policies, as shown in the following table:

To use…	Do this…
An existing policy	Select one or more policies from the list.
A new policy	1.Click ➕ **Create**. 2. Complete the New Backup Policy wizard to return to the Create Resource Group wizard.

   In Linked Mode, the list includes policies in all the linked vCenters. You must select a policy that is on the same vCenter as the resource group.

6. On the **Schedules** page, configure the backup schedule for each selected policy.



In the starting hour field, enter a time other than zero.

You must fill in each field. The SnapCenter VMware plug-in creates schedules in the time zone in which the SnapCenter VMware plug-in is deployed. You can modify the time zone by using the SnapCenter Plug-in for VMware vSphere GUI.

Modify the time zones for backups.

7. Review the summary, and then click **Finish**.

   Before you click **Finish**, you can go back to any page in the wizard and change the information.

   After you click **Finish**, the new resource group is added to the resource groups list.

> If the quiesce operation fails for any of the VMs in the backup, then the backup is marked as not VM- consistent even if the policy selected has VM consistency selected. In this case, it is possible that some of the VMs were successfully quiesced.

Manage compatibility check failures

SnapCenter performs compatibility checks when you attempt to create a resource group.

Reasons for incompatibility might be:

- VMDKs are on unsupported storage; for example, on an ONTAP system running in 7-Mode or on a non-ONTAP device.
- A datastore is on NetApp storage running Clustered Data ONTAP 8.2.1 or earlier.

  SnapCenter version 4.x supports ONTAP 8.3.1 and later.

  The SnapCenter Plug-in for VMware vSphere does not perform compatibility checks for all ONTAP versions; only for ONTAP versions 8.2.1 and earlier. Therefore, always see the NetApp Interoperability Matrix Tool (IMT) for the latest information about SnapCenter support.

- A shared PCI device is attached to a VM.
- A preferred IP is not configured in SnapCenter.
- You have not added the storage VM (SVM) management IP to SnapCenter.
- The storage VM is down.

To correct a compatibility error, perform the following:

1. Make sure the storage VM is running.
2. Make sure that the storage system on which the VMs are located have been added to the SnapCenter Plug-in for VMware vSphere inventory.
3. Make sure the storage VM is added to SnapCenter. Use the Add storage system option on the VMware vSphere web client GUI.
4. If there are spanning VMs that have VMDKs on both NetApp and non-NetApp datastores, then move the VMDKs to NetApp datastores.

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